

## 6.4 Sampling and estimation

### Probability and Statistics 2 · 2026 examination · Concept guide and original exam-style practice

#### Understand the concept

A statistic varies from sample to sample; its sampling distribution describes that variation. For independent observations from suitable population, sample mean is unbiased for  $\mu$  with  $\text{Var}(\bar{X}) = \sigma^2/n$ . A confidence interval combines estimate with critical value times standard error; distinguish confidence of method from probability of a fixed parameter.

#### Worked example: follow the reasoning

If  $\sigma=10$  and  $n=25$ , standard error of sample mean is  $10/\sqrt{25}=2$ .

#### Structured practice

Answer in the space provided. Show working and units where relevant.

1 A normal population has known  $\sigma=12$ ; a random sample  $n=36$  has mean 50. Form approximate 95% z interval for  $\mu$ . [3]

Final answer: \_\_\_\_\_

## Answers and explanation

### Question 1 [3 marks]

$SE=12/\sqrt{36}=2$  [1];  $\text{margin}=1.96\times 2=3.92$  [1]; interval  $(46.08, 53.92)$  [1].

### Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.